



MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY

ITWM106

Mobile Application Development 3 (cross-platform)



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Overview of React and React Native



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What is React?

- React is a **front-end library** for building composable user interfaces
- It is developed by Facebook.
- It is used for handling the view layer for web and mobile apps.
- ReactJS allows us to create reusable UI components.
- React can also render on the server using Node, and it can power native apps using React Native.
- It is currently one of the most popular JavaScript libraries and has a strong foundation and large community behind it.



React Features

- **JSX** – JavaScript syntax extension. It isn't necessary to use JSX in React development, but it is recommended.
- **Components** – React is all about components. You need to think of everything as a component. This will help you maintain the code when working on larger scale projects.



What is React Native?

- React Native lets you build mobile apps using only JavaScript.
- It uses the same design as React, letting you compose a rich mobile UI from declarative components.
- With React Native, you don't build a mobile web app, an HTML5 app, or a hybrid app; you build a real mobile app that's indistinguishable from an app built using Swift or Kotlin.
- React Native uses the same fundamental UI building blocks as regular iOS and Android apps.
- You just put those building blocks together using JavaScript and React.



React Native Features

- **React** – This is a Framework for building web and mobile apps using JavaScript.
- **Native** – You can use native components controlled by JavaScript.
- **Platforms** – React Native supports IOS and Android platform.



Advantages of React Native

- **JavaScript** – You can use the existing JavaScript knowledge to build native mobile apps.
- **Code sharing** – You can share most of your code on different platforms.
- **Community** – The community around React and React Native is large, and you will be able to find any answer you need.



Limitations of React Native

- **Native Components** – If you want to create native functionality which is not created yet, you will need to write some platform specific code.



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Environment Setup

- Setting up React Native environment depends on the operating system. For the updated step-by-step process on installing react native dependencies on your computer, please visit the following link:
- <https://reactnative.dev/docs/environment-setup>
 - <https://docs.expo.dev/get-started/set-up-your-environment/>



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Creating an application

- You can use React Native without a Framework, however most developers benefit from using a React Native Framework like Expo.
- Expo provides features like file-based routing, high-quality universal libraries, and the ability to write plugins that modify native code without having to manage native files.
- To initialize an application using expo, open your terminal, navigate to your desired directory and type the following:

`npx create-expo-app@latest --template blank ./`

Note: Node should be installed before you run this command

To run the program, enter the following command in terminal:

`npx expo start`



Project Structure

- App.js
 - Main entry point of the app.
 - We define your root component here.
 - Can be replaced with navigation or other layout logic.



Project Structure

- app.json or app.config.js
 - This is the project configuration file for Expo.
 - Includes metadata like the app name, slug, icons, splash screen, and permissions.



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Project Structure

- assets/
 - Contains static files like images, fonts, icons, etc.
 - Referenced in app.json or in code.



Project Structure

- package.json
 - Standard Node.js package file.
 - Includes dependencies (like react-native, expo, expo-camera, etc.) and scripts (like start, android, ios).



Project Structure

- node_modules/
 - Contains installed dependencies



Project Structure

- .gitignore
 - Tells Git to ignore node_modules, build artifacts, etc.





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